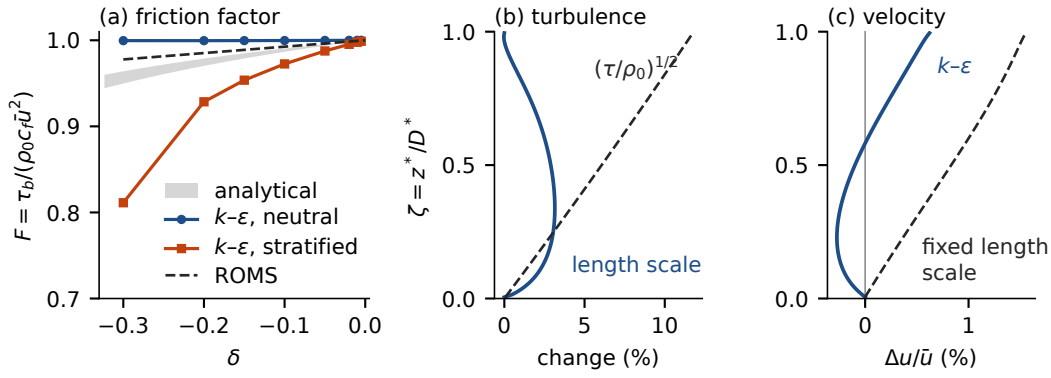
(a) friction factor
1.0
0.9
0.8
0.7
$F = \tau_b/(\rho_0 c_f \bar{u}^2)$
analytical
k–ε, neutral
k–ε, stratified
ROMS
−0.3
−0.2
−0.1
0.0
δ
(b) turbulence
1.0
0.5
0.0
$\zeta = z^*/D^*$
$(\tau/\rho_0)^{1/2}$
length scale
0
5
10
change (%)
(c) velocity
1.0
0.5
0.0
k–ε
fixed length scale
0
1
$\Delta u/\bar{u}$ (%)